Statistical foundations of business analytics

Day 2 Association rule mining

Market baskets, Apriori and recommendation decisions

27 October 2026
Lecture slides for a 150 minute session

The lecture route

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What can ShopEase learn from completed orders

Coffee orders often contain milk. ShopEase wants to use purchasing patterns to improve product recommendations.

The analyst must explain how often the combination occurs and whether it exceeds an appropriate baseline.

The final decision concerns a proposed action and its business value.

All datasets and business results in this deck are hypothetical teaching examples.

Does buying coffee cause a milk purchase

$$\{\text{Coffee}\} \Rightarrow \{\text{Milk}\}$$

The completed-order data show that the two products often appear together.

Could popularity, a breakfast promotion or customer preferences produce the same pattern?

A purchasing pattern can have several explanations

Customers may regard the products as complementary.

Milk may already be popular in almost every type of basket.

A promotion or a shared customer segment may encourage both purchases.

The rule describes cooccurrence. A randomized comparison is needed to estimate the effect of a recommendation.

Discovery validation and intervention

Business question	Appropriate evidence
Which products occur together	Historical basket counts
Does the pattern persist	Metrics in later or held-out data
Does a prompt increase business value	A suitable randomized experiment

Candidate bundles, cross-selling ideas and picking patterns can all begin with basket analysis.

One completed order defines one basket

A transaction groups products into one observation. Here, the transaction is one completed order.

An item is the product representation, such as an exact SKU or a category.

An itemset is a collection of distinct items, such as bread and milk.

The order ID identifies the basket. The customer ID identifies who placed it.

One customer places five orders

A customer places five legitimate completed orders during the month.

Two orders contain exactly the same products.

How many baskets belong in a completed-order analysis?

Five orders contribute five baskets

Each order is a separate purchasing event, even when its products match another order.

Grouping the same customer over six months answers a different question. Purchases months apart can then appear together.

Repeated orders from one customer also matter when estimating uncertainty or generalizing to new customers.

Three units of milk still mean milk is present once

Recorded purchase	Binary basket representation
Three units of milk	Milk = 1
Bread and two milk line items	Bread = 1, Milk = 1

Quantity remains useful for revenue and inventory. Basic basket presence records whether a product appears in that order.

A pair containing bread and milk requires both products, regardless of their quantities.

Order lines become sets of products

Order ID	Raw product lines	Resulting itemset
T1	Bread, Milk, Milk, Butter	Bread, Milk, Butter
T6	Bread, Milk, Eggs	Bread, Milk, Eggs
T7	Bread, Butter	Bread, Butter

Normalize product IDs, then record one presence per order and product.

A binary basket by item matrix

Order	Bread	Milk	Butter	Eggs
T1	1	1	1	0
T6	1	1	0	1
T7	1	0	1	0
T9	0	0	0	1

A one means present. A zero means known absence from the recorded basket.

Basket construction needs consistent records

Choose the order status: placed, paid, fulfilled or retained after returns.

Remove duplicate records of the same order and product. Check whether a join multiplied order lines.

Keep distinct legitimate orders with identical contents.

Retain a stable total basket count, including baskets that lack the products currently being investigated.

The population behind every reported rule

Definition	Example choice
Transaction	One completed order
Item	A consistently mapped product category
Population	Eligible online grocery orders
Period	A specified month in one time zone
Presence	Purchased at least once within that order

Incomplete records need investigation

If a complete order lists bread and butter, milk can be recorded as absent.

If half the order lines failed to load, an apparent absence of milk may be a data problem.

The item level also changes the analysis. All milk variants combined form a different item from a particular milk SKU.

Reliable comparisons require consistent extraction and product mapping.

A separate dataset with 100 baskets

Observed quantity	Count
All completed baskets	100
Baskets containing coffee	40
Baskets containing milk	50
Baskets containing both	30

The coffee and milk counts include baskets that contain both products.

The counts determine a complete two by two table

Basket group	Milk present	Milk absent	Total
Coffee present	30	?	40
Coffee absent	?	?	60
Total	50	50	100

How many baskets contain coffee without milk, milk without coffee, and neither product?

Ten coffee baskets contain no milk

Basket group	Milk present	Milk absent	Total
Coffee present	30	10	40
Coffee absent	20	40	60
Total	50	50	100

Coffee only = 40 - 30 = 10. Milk only = 50 - 30 = 20. Neither = 100 - 30 - 10 - 20 = 40.

Support uses all baskets as the denominator

$$s(\text{Coffee}) = \frac{40}{100} = 40\%, \quad s(\text{Milk}) = \frac{50}{100} = 50\%$$

$$s(\text{Coffee and Milk}) = \frac{30}{100} = 30\%$$

The support of Coffee to Milk is the joint support: 30 baskets out of 100.

Support describes how common the required combination is in the whole population.

Could 60 baskets contain both products

An analyst reports 100 baskets, 40 containing coffee, and 60 containing both coffee and milk.

Can these counts describe the same valid dataset?

Every joint basket is also a coffee basket

$$c(\text{Coffee and Milk}) \leq c(\text{Coffee})$$

A basket containing both products must belong to the coffee group.

The joint count cannot exceed 40 when only 40 baskets contain coffee.

This subset relationship later explains why Apriori can discard some larger combinations.

Confidence changes the denominator

For Coffee to Milk, consider only the 40 baskets containing coffee.

$$\text{conf}(C \Rightarrow M) = \frac{30}{40} = 75\%$$

Among those coffee baskets, 75% also contain milk.

This describes completed baskets. It does not estimate the response rate to a future milk prompt.

The conditional probability identity

$$P(M \mid C) = \frac{P(C \cap M)}{P(C)}$$

$$\frac{30/100}{40/100} = \frac{30}{40} = 0.75$$

The total basket count cancels. The remaining denominator is the number of coffee baskets.

The reverse rule uses milk as its starting group

Coffee appears in 40 baskets, milk in 50, and both in 30.

$$\text{conf}(C \Rightarrow M) = \frac{30}{40} = 75\%$$

What is the confidence of Milk to Coffee? Does joint support change?

Reverse confidence is 60 percent

$$\text{conf}(M \Rightarrow C) = \frac{30}{50} = 60\%$$

Direction	Joint support	Confidence
Coffee to Milk	30%	75%
Milk to Coffee	30%	60%

The joint count stays 30. The antecedent group changes from 40 to 50 baskets.

The question determines the denominator

Quantity	Numerator	Denominator
Coffee support	40 coffee baskets	100 baskets
Coffee and milk support	30 joint baskets	100 baskets
Coffee to milk confidence	30 joint baskets	40 coffee baskets
Milk to coffee confidence	30 joint baskets	50 milk baskets

A confidence requires at least one observed antecedent basket.

A rule has 95 percent confidence

A report says that a rule has 95% confidence.

Does this mean a 95% confidence interval or a 95% probability that the rule is correct?

The term describes a conditional frequency

A 95% association-rule confidence means that 95% of the observed antecedent baskets also contain the consequent.

The underlying counts could be 19 of 20 or 1,900 of 2,000. Those counts have different implications for precision.

A statistical confidence level and a probability assigned to a hypothesis are different concepts.

Lift compares confidence with a baseline

Milk appears in 75% of coffee baskets and 50% of all baskets.

$$\text{lift}(C \Rightarrow M) = \frac{P(M \mid C)}{P(M)} = \frac{0.75}{0.50} = 1.5$$

Milk is observed 1.5 times as frequently among coffee baskets as in the overall population.

That comparison is a 50% relative increase in frequency, or a 25 percentage point difference.

Independence supplies the comparison baseline

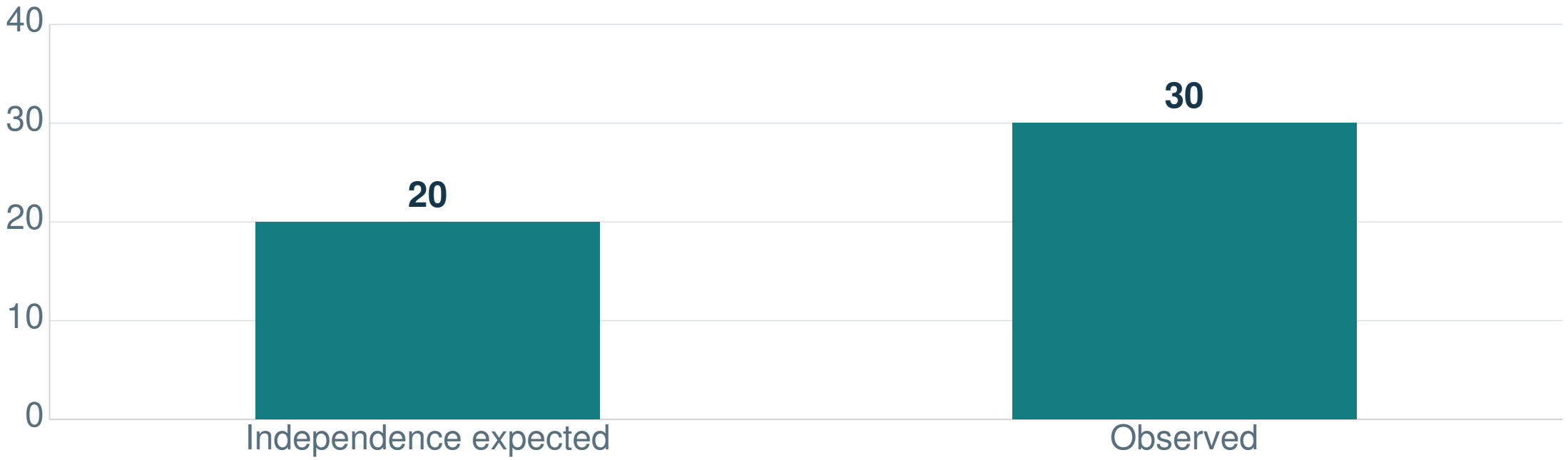
$$P(C \cap M) = P(C)P(M) \quad \text{under independence}$$

$$100(0.40)(0.50) = 20 \text{ expected joint baskets}$$

$$\text{lift} = \frac{\text{observed joint count}}{\text{independence expected count}} = \frac{30}{20} = 1.5$$

The expected count is a statistical baseline, rather than a randomized control arm.

Thirty joint baskets compared with twenty expected



The ratio is 1.5. The excess is ten joint baskets in this historical population.

Neither quantity establishes the effect of displaying a recommendation.

Reversing a rule leaves its standard lift unchanged

$$\text{lift}(C \Rightarrow M) = \frac{P(C \cap M)}{P(C)P(M)}$$

$$\text{lift}(M \Rightarrow C) = \frac{0.60}{0.40} = 1.5$$

The symmetric formula uses the same joint support and the same two marginal supports.

Confidence can change with direction. Support and standard lift stay the same for the same two itemsets and population.

Interpreting lift relative to one

Lift	Description in the chosen population
Greater than 1	More joint occurrence than independence predicts
Equal to 1	Observed proportions match that baseline
Less than 1	Less joint occurrence than independence predicts

A value below one does not establish product substitution. Availability, promotions and customer mix can produce such a pattern.

Milk appears in 90 percent of all baskets

In a different population, 80% of coffee baskets contain milk. Milk appears in 90% of all baskets.

Is milk unusually common among coffee baskets? Calculate lift.

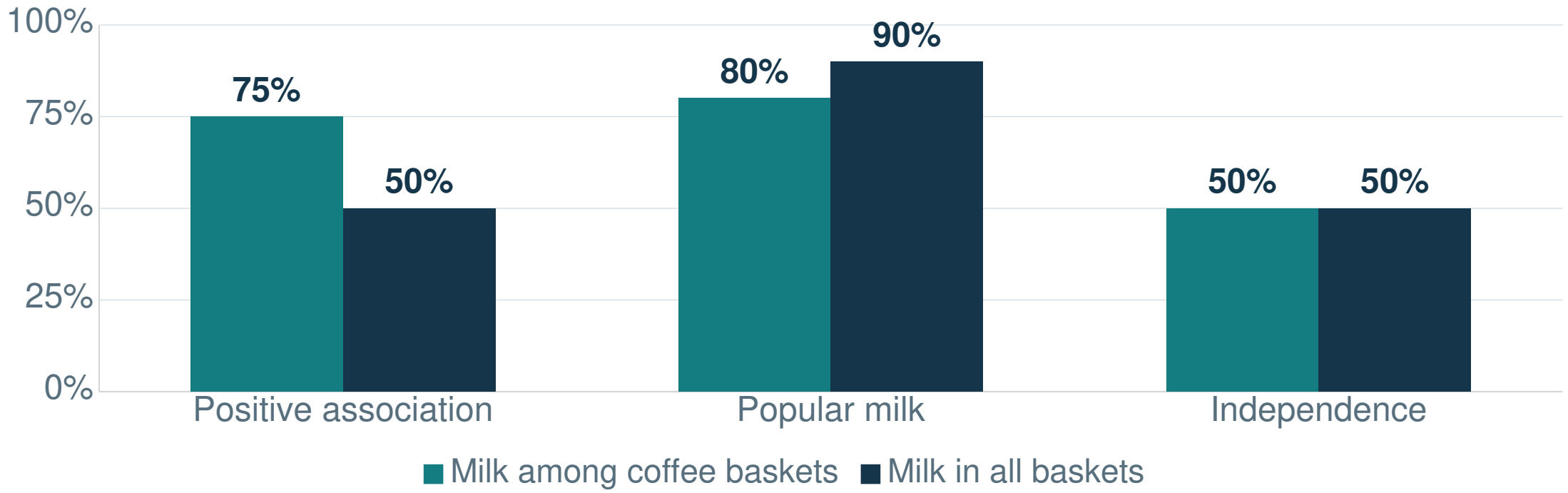
High confidence can coexist with lift below one

$$\text{lift} = \frac{0.80}{0.90} \approx 0.889$$

Milk is common among coffee baskets, but it is even more common in the population overall.

Confidence alone would hide that comparison. The 90% baseline changes the interpretation of the 80% confidence.

Three separate populations need their own baselines



The respective lifts are 1.50, 0.889 and 1.00. These rows use different hypothetical datasets.

An analyst keeps only the coffee baskets

Return to the original 100 baskets: 40 contain coffee, 50 contain milk, and 30 contain both.

The analyst deletes all 60 baskets without coffee and recalculates lift.

What are coffee support, milk support and lift in this restricted population?

Filtering changes the analysis population

$$s(C) = \frac{40}{40} = 1, \quad s(M) = \frac{30}{40} = 0.75, \quad s(CM) = 0.75$$

$$\text{lift} = \frac{0.75}{1(0.75)} = 1$$

Every remaining basket contains coffee, so coffee presence no longer distinguishes groups.

The original lift of 1.5 used all 100 baskets. The restricted result answers a different question.

Five minute break

The next problem is discovering useful patterns when the product catalogue is large.

Support will determine which combinations qualify. Confidence and lift will evaluate the resulting rules.

From individual rules to a search problem

Evaluating Coffee to Milk begins with a chosen rule. Mining begins with a catalogue and many possible combinations.

The search must discover combinations that occur often enough, then determine which directions satisfy the chosen criteria.

A combination and a directional rule are different objects.

Three products produce seven nonempty itemsets

Size	Possible itemsets	Count
One item	Bread, Milk, Butter individually	3
Two items	Bread + Milk, Bread + Butter, Milk + Butter	3
Three items	Bread + Milk + Butter	1

Bread + Milk is the same itemset as Milk + Bread. Itemset order does not matter.

Each additional product doubles the possibilities

For each product, a candidate combination either includes it or leaves it out.

$$2 \times 2 \times 2 = 2^3 = 8 \quad \text{including the empty set}$$

$$2^3 - 1 = 7, \qquad 2^{10} - 1 = 1023, \qquad 2^{20} - 1 = 1{,}048{,}575$$

These are possible nonempty itemsets. Some combinations may never occur in the observed baskets.

Two stages keep support and confidence distinct

Stage	Question	Criterion
Frequent itemsets	Does the combination occur often enough	Minimum support
Rule generation	Which directed splits qualify	Confidence, then other filters

In Apriori notation, Ck contains candidate itemsets of size k. Lk retains the frequent ones.

Business usefulness remains a separate judgment.

A support threshold becomes a count threshold

$$\text{Minimum count} = \lceil N \times s_{\min} \rceil$$

$$N = 10, \quad s_{\min} = 0.30 \quad \Longrightarrow \quad \lceil 10(0.30) \rceil = 3$$

At least three baskets must contain every required item.

Equality passes. A count of three meets a minimum count of three.

The 30% support, 60% confidence and 1.05 lift thresholds used here are teaching choices.

Ten baskets for the Apriori calculation

Basket ID	Products in the completed order
T1	Bread, Milk, Butter
T2	Bread, Milk, Butter
T3	Bread, Milk
T4	Bread, Milk, Butter
T5	Bread, Milk
T6	Bread, Milk, Eggs
T7	Bread, Butter
T8	Milk, Butter
T9	Eggs
T10	Eggs

T1 T2 and T4 contain the same products

Order ID	Products
T1	Bread, Milk, Butter
T2	Bread, Milk, Butter
T4	Bread, Milk, Butter

Should these three rows collapse into one basket?

Identical contents can occur in separate orders

T1, T2 and T4 identify three legitimate purchasing events.

The Bread + Milk + Butter combination occurred three times. Repetition is the frequency we want to measure.

Remove accidental duplicate records of the same order. Preserve different orders that happen to contain the same products.

The same customer could also place more than one legitimate order.

First count individual products

Product	Basket IDs	Count	Support
B bread	T1 T2 T3 T4 T5 T6 T7	7	70%
M milk	T1 T2 T3 T4 T5 T6 T8	7	70%
U butter	T1 T2 T4 T7 T8	5	50%
E eggs	T6 T9 T10	3	30%

All four products meet the minimum count of three. U represents butter so every product has a distinct code.

Individual popularity does not determine pair counts

Bread and milk each occur seven times. Butter occurs five times and eggs three times.

The six candidate pairs are BM, BU, MU, BE, ME and UE.

Which pairs appear in at least three of the ten baskets?

Three pairs meet the support threshold

Candidate pair	Count	Support	Retain
BM	6	60%	Yes
BU	4	40%	Yes
MU	4	40%	Yes
BE	1	10%	No
ME	1	10%	No
UE	0	0%	No

BM, BU and MU pass. Every pair containing eggs fails.

Adding product requirements cannot increase support

Every basket containing bread, milk and eggs must also contain bread and eggs.

$$c(\{B, M, E\}) \leq c(\{B, E\})$$

Bread + Eggs occurs once. A larger combination requiring both cannot occur more than once.

The same argument applies to any subset and larger itemset in the same population.

Apriori discards impossible frequent extensions

Candidate triple	A required pair	Reason
BME	BE has count 1	Cannot reach count 3
BUE	BE has count 1	Cannot reach count 3
MUE	ME has count 1	Cannot reach count 3
BMU	BM, BU, MU all frequent	Needs its own count

Pruning avoids counting candidates already ruled out by the subset relationship.

All three pairs of a triple are frequent

Suppose BM, BU and MU each pass the minimum support threshold.

Must BMU also pass?

Consider whether the three pairs might occur in different baskets.

Frequent subsets allow a candidate but do not prove it

Some baskets can contain bread and milk, others bread and butter, and others milk and butter.

Those pair counts can be large even when few baskets contain all three products.

A frequent triple requires frequent pairs. The reverse implication does not hold.

The remaining triple must be counted directly.

The triple occurs in T1 T2 and T4

Basket	Contains bread	Contains milk	Contains butter
T1	Yes	Yes	Yes
T2	Yes	Yes	Yes
T4	Yes	Yes	Yes

$$c(BMU) = 3, \qquad s(BMU) = \frac{3}{10} = 30\%$$

BMU is frequent. The four-item set requires failed egg pairs, so the search stops.

The union combines lists of required products

$$A = \{\text{Bread, Milk}\}, \qquad B = \{\text{Butter}\}$$

$$A \cup B = \{\text{Bread, Milk, Butter}\}$$

A basket must contain every product on the combined list.

In event notation, the event of containing A and the event of containing B occur together.

A and B on this slide label whole itemsets.

Where the counts six and three come from

Basket containing bread and milk	Also contains butter
T1	Yes
T2	Yes
T3	No
T4	Yes
T5	No
T6	No

$$c(A) = 6, \qquad c(A \cup B) = 3$$

Three of the six bread and milk baskets have butter

$$\text{conf}(A \Rightarrow B) = \frac{c(A \cup B)}{c(A)} = \frac{3}{6} = 50\%$$

$$s(A \Rightarrow B) = \frac{c(A \cup B)}{N} = \frac{3}{10} = 30\%$$

Confidence uses the six qualifying antecedent baskets. Support uses all ten baskets.

One three-product itemset can produce several rules

Take the frequent itemset Bread + Milk + Butter.

Assign each product to the left side or the right side of a rule, keeping both sides nonempty.

How many directed splits are possible?

Three products produce six directed splits

$$2^3 - 2 = 6$$

There are two choices of side for each product, giving eight assignments.

Exclude the assignment with everything on the left and the assignment with everything on the right.

For a k-itemset, the number of nonempty directed splits is 2 to the power k, minus 2.

Six rules use all three products

Rule	Required consequent
BM to U	Butter
BU to M	Milk
MU to B	Bread
B to MU	Both milk and butter
M to BU	Both bread and butter
U to BM	Both bread and milk

B to M comes from the separate pair BM. It does not use all three products.

The joint count stays three but the denominator changes

Antecedent	Count
BM bread and milk	6
B bread	7
U butter	5

Find confidence for BM to U, B to MU, and U to BM. Which meet 60%?

A frequent itemset does not guarantee high confidence

Rule	Antecedent count	Confidence	Meets 60%
BM to U	6	3 / 6 = 50%	No
BU to M	4	3 / 4 = 75%	Yes
MU to B	4	3 / 4 = 75%	Yes
B to MU	7	3 / 7 = 42.9%	No
M to BU	7	3 / 7 = 42.9%	No
U to BM	5	3 / 5 = 60%	Yes

Every row has joint support 3 / 10 = 30%.

The frequent pairs also generate rule directions

Rule	Confidence	Consequent baseline	Lift
B to M	6 / 7 = 85.7%	70%	1.224
M to B	6 / 7 = 85.7%	70%	1.224
B to U	4 / 7 = 57.1%	50%	1.143
U to B	4 / 5 = 80%	70%	1.143
M to U	4 / 7 = 57.1%	50%	1.143
U to M	4 / 5 = 80%	70%	1.143

B to U and M to U fail the chosen 60% confidence threshold.

Seven rules pass the confidence threshold

Rule	Support	Confidence	Lift
B to M	60%	85.7%	1.224
M to B	60%	85.7%	1.224
U to B	40%	80%	1.143
U to M	40%	80%	1.143
BU to M	30%	75%	1.071
MU to B	30%	75%	1.071
U to BM	30%	60%	1.000

Displayed values are rounded. Apply thresholds to the unrounded metrics.

A lift threshold of 1.05 leaves six rules

U to BM has confidence 60%, but bread and milk already occur together in 60% of all baskets.

$$\text{lift}(U \Rightarrow BM) = \frac{0.60}{0.60} = 1$$

$$\text{lift}(BU \Rightarrow M) = \frac{0.75}{0.70} \approx 1.071$$

B to M, M to B, U to B, U to M, BU to M and MU to B pass all three chosen filters.

Some systems restrict the consequent to one product, reducing the usable rule set.

Does adding butter strengthen the milk prediction

Antecedent	Milk confidence
Bread	85.7%
Bread and butter	75%

Does the more detailed antecedent necessarily make a stronger rule?

More conditions can reduce confidence and reach

In these baskets, the bread-and-butter group has a lower milk frequency than the broader bread group.

The more complex rule also applies to fewer baskets and needs sufficient validation data.

Support-based subset pruning is safe. Confidence and lift do not have the same general monotonicity.

The two sides of a rule must be disjoint. Bread to Bread would simply repeat the same requirement.

Thresholds determine which patterns enter the shortlist

At 30% minimum support, BM, BU and MU are frequent pairs in the ten-basket dataset.

Raising minimum support to 50% leaves BM as the only frequent pair. The triple BMU also fails.

Low thresholds increase the search and can retain unstable patterns. High thresholds can exclude useful niche combinations.

Choose thresholds with business reach, product rarity and available evidence in mind.

A very large lift can come from very few baskets

A separate population contains 10,000 completed baskets.

Quantity	X to Y	R to S
Antecedent count	10	2,000
Consequent count	20	1,500
Joint count	5	600

Calculate confidence and lift. Which rule has the larger observed joint count?

Lift 250 rests on five observed joint baskets

Quantity	X to Y	R to S
Support	0.05%	6%
Confidence	5 / 10 = 50%	600 / 2,000 = 30%
Consequent baseline	0.2%	15%
Lift	250	2
Observed joint baskets	5	600

A rare pattern may matter, but its size, repeatability and business reach need evidence.

Shared shopping-trip types can create pooled association

This separate dataset splits 100 baskets by trip type.

Trip type	Baskets	Coffee	Milk	Both
Breakfast	50	40	40	32
Other	50	10	10	2
Pooled	100	50	50	34

Find lift within each segment and in the pooled data.

Each segment has lift one while pooled lift is 1.36

$$\text{Breakfast lift} = \frac{32/50}{(40/50)(40/50)} = 1$$

$$\text{Other lift} = \frac{2/50}{(10/50)(10/50)} = 1$$

$$\text{Pooled lift} = \frac{34/100}{(50/100)(50/100)} = 1.36$$

The pooled association reflects the mix of trip types. It remains a description of the pooled data, not a causal effect.

A promising historical pattern needs later validation

Discover rules in earlier data and freeze the initial shortlist. Recalculate counts and metrics in a later period.

Check stock availability, promotions and whether the customer mix changed.

Repeated tuning on the validation period requires a further untouched test period.

Searching many candidates can produce impressive-looking rules by chance. Lift itself is not a p-value.

Counts and generalization matter alongside ratios

A complete dataset has definite observed counts and supports. Claims about future traffic introduce uncertainty and assumptions.

A confidence of 100% may come from only a handful of antecedent baskets.

Several orders from one customer can be dependent. Hold out customers when the target is new customers.

For future traffic, use a suitable time split and account for customer overlap when estimating uncertainty.

Only ten historical coffee baskets lack milk

Original coffee dataset	Count
Coffee baskets	40
Coffee baskets already containing milk	30
Coffee baskets without milk	10

The 75% confidence includes customers who purchased milk without the proposed prompt.

The ten historical baskets suggest a possible opportunity. They do not measure future prompt response.

A recommendation needs a valid serving trigger

At the moment of display, coffee is in the cart, milk is absent, milk is available and the customer is eligible.

Products already present should be excluded from the recommendation.

A completed basket does not establish which item entered first or what the customer intended before a prompt.

The data available at serving time determine whether the rule can support the proposed action.

The historical rule becomes an experiment hypothesis

Protocol field	Example
Population	Customers meeting the live trigger
Assignment unit	Account at its first eligible opportunity
Treatment	A specified milk recommendation prompt
Control	Existing experience
Primary business outcome	Net contribution per assigned eligible customer

Incremental response comes from the randomized comparison

$$\hat{\Delta} = \hat{p}_{\text{prompt}} - \hat{p}_{\text{control}}$$

Keep assignment persistent and allow all assigned customers to finish the agreed outcome window.

Repeated prompt impressions do not create independent customer trials.

Evaluate contribution after serving costs, discounts and displaced purchases. A historical lift of 1.5 does not imply a 50% prompt effect.

Worked exercises

Case	Focus
Pasta, sauce and cheese	Counts, rule metrics and Apriori
Two candidate rules	Lift, reach and historical opportunities
A randomized recommendation	Incremental purchases and net contribution

Each case uses its own labelled dataset.

Eight completed baskets for the first two exercises

Order	Products
T1	Pasta, Sauce, Cheese
T2	Pasta, Sauce
T3	Pasta, Sauce, Cheese
T4	Pasta, Sauce
T5	Pasta, Cheese
T6	Sauce, Cheese
T7	Pasta, Sauce
T8	Cheese

Exercise 1 on pasta to sauce

Count pasta baskets, sauce baskets, cheese baskets and baskets containing pasta with sauce.

Calculate support, confidence and lift for Pasta to Sauce.

P, S and C denote pasta, sauce and cheese.

Pasta to sauce has support 62.5 percent

Count	Pasta	Sauce	Cheese	Pasta + Sauce
Baskets	6	6	5	5

$$s(P \Rightarrow S) = \frac{5}{8} = 62.5\%$$

$$\text{conf}(P \Rightarrow S) = \frac{5}{6} = 83.3\%$$

$$\text{lift}(P \Rightarrow S) = \frac{5/6}{6/8} = \frac{10}{9} \approx 1.111$$

Exercise 1 on the reverse rule and cheese

Pasta and sauce each occur in six baskets. Their joint count is five.

Cheese occurs in five baskets and pasta with cheese occurs in three.

Find confidence for Sauce to Pasta. Then find confidence and lift for Cheese to Pasta.

Cheese to pasta has lift below one

$$\text{conf}(S \Rightarrow P) = \frac{5}{6} = 83.3\%$$

$$\text{conf}(C \Rightarrow P) = \frac{3}{5} = 60\%$$

$$\text{lift}(C \Rightarrow P) = \frac{0.60}{6/8} = 0.80$$

Pasta is less common among cheese baskets than in the population overall. Equal reverse confidence here results from equal pasta and sauce counts.

Exercise 2 on frequent pairs and the triple

Use the eight pasta, sauce and cheese baskets.

Minimum support count is three baskets, equivalent to support of 3 / 8 = 37.5%. Equality passes.

Which singletons and pairs qualify? Count PSC and determine whether the triple is frequent.

Every pair is frequent but the triple fails

Size	Candidate counts	Frequent itemsets
One item	P = 6, S = 6, C = 5	P, S, C
Two items	PS = 5, PC = 3, SC = 3	PS, PC, SC
Three items	PSC = 2	None

PSC appears only in T1 and T3. Frequent pairs are necessary for a frequent triple, but are not sufficient.

Exercise 2 on the final rule shortlist

The frequent pairs are PS, PC and SC. The triple is infrequent.

Require confidence of at least 70% and lift of at least 1.05.

Which directed rules pass both filters?

Only pasta to sauce and sauce to pasta qualify

Rule	Confidence	Lift	Retain
P to S	83.3%	1.111	Yes
S to P	83.3%	1.111	Yes
P to C	50%	0.80	No
S to C	50%	0.80	No
C to P	60%	0.80	No
C to S	60%	0.80	No

Cheese is frequent by itself, yet has negative relative association with pasta and sauce here.

Exercise 3 on rule ranking and potential reach

A different dataset contains 10,000 completed baskets.

Quantity	X to Y	R to S
Antecedent count	50	2,000
Consequent count	100	1,500
Joint count	20	600

Calculate support, confidence, lift and the number of antecedent baskets lacking the consequent.

The larger lift has the smaller historical reach

Quantity	X to Y	R to S
Support	0.2%	6%
Confidence	40%	30%
Lift	40	2
Joint baskets	20	600
Antecedent without consequent	30	1,400

Margin, validation stability and serving eligibility also matter. These historical counts do not estimate response to a prompt.

Exercise 4 on a randomized recommendation

Arm	Assigned customers	Product purchasers
Control	500	60
Prompt	500	90

Checks pass and follow-up is complete. Each purchaser buys one unit with INR 80 contribution. The prompt costs INR 2 for every assigned treatment customer, with no extra control cost or other business effects.

Find the incremental purchase probability and net contribution per assigned customer in each arm.

The estimated purchase increase is six percentage points

$$\hat{p}_C = \frac{60}{500} = 0.12, \qquad \hat{p}_T = \frac{90}{500} = 0.18$$

$$\hat{\Delta} = 0.18 - 0.12 = 0.06 = 6 \text{ percentage points}$$

$$\text{Relative improvement} = \frac{0.06}{0.12} = 50\%$$

This compares randomized treatment and control arms. Its meaning differs from historical association-rule lift.

The estimated net gain is INR 2.80 per customer

Quantity	Control	Prompt
Contribution before serving cost	INR 4,800	INR 7,200
Serving cost	INR 0	INR 1,000
Net contribution	INR 4,800	INR 6,200
Net / assigned customer	INR 9.60	INR 12.40

$$12.40 - 9.60 = 2.80 \text{ INR per assigned customer}$$

Exercise 4 on uncertainty in the economic result

The supplied approximate 95% interval for the purchase probability difference is 0.01589 to 0.10411.

$$\Delta_{\text{net}} = 80\Delta_{\text{purchase}} - 2$$

Under the fixed contribution and cost assumptions, what is the interval for net contribution? What can the business conclude?

The net contribution interval includes a small loss

$$\text{Lower} = 80(0.01589) - 2 \approx -0.73 \text{ INR}$$

$$\text{Upper} = 80(0.10411) - 2 \approx 6.33 \text{ INR}$$

The approximate 95% net interval is -INR 0.73 to +INR 6.33 per assigned customer.

The purchase interval is positive, while serving costs shift the economic interval downward.

The business recommendation separates the two outcomes

The planned analysis supports increased product purchasing: +6 percentage points, with an approximate interval of +1.59 to +10.41 points.

Estimated net contribution rises by INR 2.80 per assigned customer, with an interval from -INR 0.73 to +INR 6.33.

These data do not yet establish a positive net contribution effect at the conventional two-sided 5% level.

Use the prespecified economic criteria and uncertainty tolerance. Actual variable margins and displaced purchases need the full account-level business outcome.

Five review questions

1 What denominator belongs in support and in confidence?

2 Can confidence be 80% while lift is below one?

3 What larger itemsets can Apriori safely discard?

4 Why do T1, T2 and T4 remain three observations?

5 What evidence estimates the effect of a recommendation?

The distinctions needed for a defensible analysis

1 Support uses all baskets. Confidence uses antecedent baskets.

2 Yes, when the consequent is even more common overall.

3 Every superset of an itemset below the support threshold.

4 They are separate legitimate orders with identical contents.

5 A suitable randomized action-versus-control comparison, with business outcomes and uncertainty.

Additional example on phones and protective cases

Observed quantity	Baskets
All completed baskets	500
Contain a phone	100
Contain a protective case	150
Contain both	60

Find support, confidence and lift for Phone to Case. Then reverse the rule.

The directions share lift two but have different confidence

Rule	Support	Confidence	Baseline	Lift
Phone to Case	12%	60%	30%	2
Case to Phone	12%	40%	20%	2

$$\text{Expected joint count} = 500(0.20)(0.30) = 30, \qquad \frac{60}{30} = 2$$

Historical cooccurrence can suggest a case recommendation. Its incremental purchase effect still requires an experiment.

Optional measure on absolute excess cooccurrence

$$\text{leverage} = P(C \cap M) - P(C)P(M)$$

$$0.30 - (0.40)(0.50) = 0.10$$

In 100 baskets, the observed joint count exceeds the independence expected count by ten.

Lift expresses a ratio of 1.5. Leverage expresses an absolute support difference of 0.10.

Neither measures purchases caused by a proposed prompt.

Optional comparison with noncoffee baskets

Group	Milk baskets	All baskets	Milk frequency
Coffee	30	40	75%
Noncoffee	20	60	33.3%
All	50	100	50%

$$75\% - 33.3\% \approx 41.7 \text{ points}, \quad 75\% - 50\% = 25 \text{ points}$$

The reference groups differ. The overall population includes the coffee group.

Optional uncertainty in a conditional frequency

In the coffee example, association-rule confidence is 30 / 40 = 75%.

$$\text{Wilson } 95\% \text{ interval for } 30/40 \approx [0.598,\ 0.858]$$

This interval concerns the conditional milk frequency among comparable coffee baskets.

It assumes independent comparable observations. Repeated customers can require an analysis that accounts for dependence.

An interval for lift must also account for the estimated baseline and joint structure.

Some rules follow automatically from representation

A product is mapped to its dairy category by definition. A rule from that product to Dairy repeats the mapping.

A predefined bundle can force two products into the same recorded order.

Out-of-stock products cannot appear in purchasing baskets, even when customers would prefer them.

Product definitions, promotions and availability help explain what a discovered pattern means.

Optional spreadsheet setup

Column	Contents	Rows 2 to 11
A	Order identifier	T1 to T10
B	Bread	Numeric 0 or 1
C	Milk	Numeric 0 or 1
D	Butter	Numeric 0 or 1
E	Eggs	Numeric 0 or 1

Use the ten-basket dataset. Keep the two eggs-only orders and enter headings in row 1.

Optional spreadsheet formulas for counts

Cell	Formula	Meaning
H2	=COUNTA(A2:A11)	Total baskets
H3	=SUM(B2:B11)	Bread count
H4	=SUM(C2:C11)	Milk count
H5	=SUM(D2:D11)	Butter count
H6	=COUNTIFS(B2:B11,1,C2:C11,1)	Bread and milk
H7	=COUNTIFS(B2:B11,1,D2:D11,1)	Bread and butter
H8	=COUNTIFS(C2:C11,1,D2:D11,1)	Milk and butter
H9	=COUNTIFS(B2:B11,1,C2:C11,1,D2:D11,1)	All three

Optional spreadsheet formulas for metrics

Cell	Formula	Meaning
H12	=H6/H2	Bread and milk support
H13	=H6/H3	Bread to milk confidence
H14	=H4/H2	Milk baseline
H15	=H13/H14	Bread to milk lift
H16	=H6/H4	Reverse confidence
H17	=H12-(H3/H2)*(H4/H2)	Bread and milk leverage
H18	=H9/H7	Bread and butter to milk confidence
H19	=H18/H14	Bread and butter to milk lift

The three main formulas in count notation

$$\text{support}(A \Rightarrow B) = \frac{n_{AB}}{N}$$

$$\text{confidence}(A \Rightarrow B) = \frac{n_{AB}}{n_A}$$

$$\text{lift}(A \Rightarrow B) = \frac{N n_{AB}}{n_A n_B}$$

A and B are nonempty disjoint itemsets. Confidence requires nA > 0, and lift also requires nB > 0.

Source notes and method references

Day 2 Detailed Teaching Notes, 25 pages. Slide footers connect each topic to the original notes. Cases and numerical exercises are hypothetical.

Agrawal and Srikant, Fast Algorithms for Mining Association Rules, VLDB 1994
https://rsrikant.com/papers/vldb94.pdf

Oracle Data Mining Concepts, Apriori
https://docs.oracle.com/en/database/oracle/oracle-database/18/dmcon/apriori.html

The additional phone-and-case example uses new hypothetical counts and the same metric definitions.

Data representation and implementation references

Apache MADlib, Apriori Algorithm, version 1.14 documentation
https://madlib.apache.org/docs/v1.14/group__grp__assoc__rules.html

Data Mining Group, PMML 4.4.1 Association Rules
https://dmg.org/pmml/v4-4-1/AssociationRules.html

Product definitions, the transaction universe and the observation period must remain explicit when comparing software outputs.